# Python Activity Record

**Railway Fare and Reservation Calculator**

## 1. Problem Statement

Develop a Python program to calculate a passenger's railway fare using travel distance, passenger age, and travel class. The program should calculate the base fare, apply a 20% concession for passengers aged 60 or above, add the reservation fee for the selected class, and display the final fare. It should display an invalid-input message for an invalid distance, age, or travel class.

## 2. Objectives

- To accept passenger name, distance, age, and travel class.
- To validate distance, age, and travel class.
- To calculate the base fare using distance and class rate.
- To apply a 20% concession when the passenger is 60 years or older.
- To add the reservation fee and display the final fare.

## 3. Assumptions

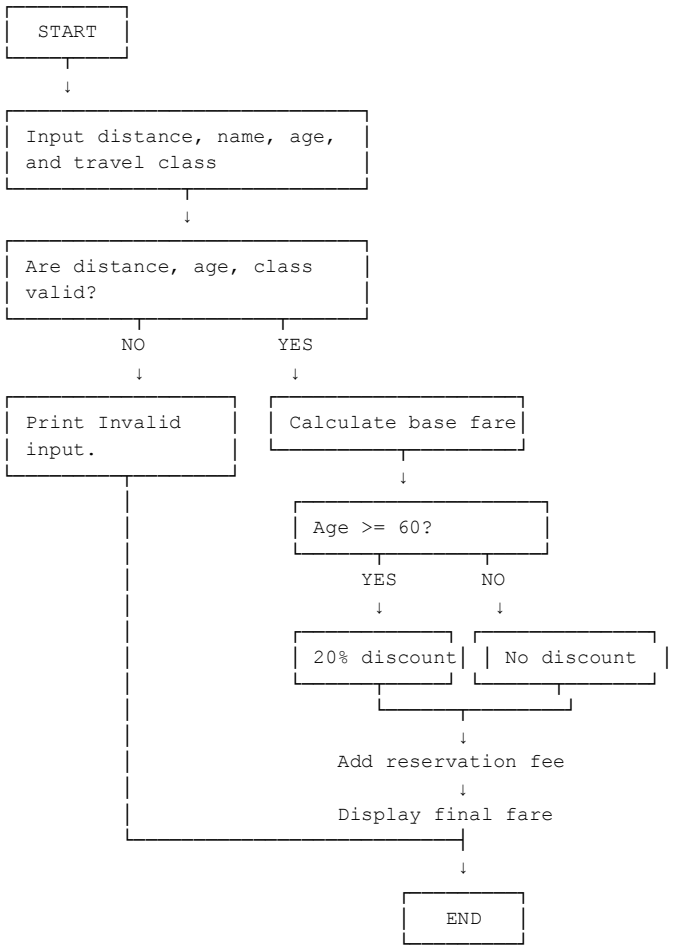
- Distance is entered as a positive number.
- Age is entered as a non-negative integer.
- Travel class must be sleeper, ac3, or ac2.
- The rate multipliers are sleeper = 1, ac3 = 2, and ac2 = 3.
- Reservation fees are sleeper = Rs. 20, ac3 = Rs. 40, and ac2 = Rs. 50.
- Passengers aged 60 or above receive a 20% concession on the base fare only.
- The code expects numeric values for distance and age; non-numeric inputs are not handled.

## 4. Algorithm

1. Start.
2. Input distance, passenger name, age, and travel class.
3. Define the rate multipliers and reservation fees for each class.
4. Check whether distance is non-positive, age is negative, or class is invalid.
5. If any input is invalid, print “Invalid input.”
6. Otherwise, calculate base fare = distance × 1.2 × class rate.
7. If age is 60 or above, calculate concession = 20% of base fare; otherwise set concession to zero.
8. Calculate final fare = base fare − concession + reservation fee.
9. Display passenger name, base fare, concession, reservation fee, and final fare.

10. Stop.

## 5. Flowchart




## 6. Pseudocode

```
BEGIN
    INPUT distance, name, age, travel_class
    rates = {sleeper: 1, ac3: 2, ac2: 3}
    fees = {sleeper: 20, ac3: 40, ac2: 50}

    IF distance <= 0 OR age < 0 OR travel_class not in rates THEN
        PRINT "Invalid input."
    ELSE
        base = distance * 1.2 * rates[travel_class]
        IF age >= 60 THEN
            concession = base * 0.20
        ELSE
            concession = 0
        END IF
        total = base - concession + fees[travel_class]
        PRINT name, base, concession, reservation fee, total
```

```
    END IF
END
```

## 7. Python Source Code

```
distance = float(input("Distance in km: "))

name = input("Passenger name: ")

age = int(input("Age: "))

travel_class = input("Class (sleeper/ac3/ac2): ").lower()

rates = {"sleeper": 1, "ac3": 2, "ac2": 3}

fees = {"sleeper": 20, "ac3": 40, "ac2": 50}

if distance <= 0 or age < 0 or travel_class not in rates:

    print("Invalid input.")

else:

    base = distance * 1.2 * rates[travel_class]

    concession = base * 0.20 if age >= 60 else 0

    total = base - concession + fees[travel_class]

    print("Passenger:", name)

    print("Base fare: Rs.", round(base, 2))

    print("Concession: Rs.", round(concession, 2))

    print("Reservation: Rs.", fees[travel_class])

    print("Final fare: Rs.", round(total, 2))
```

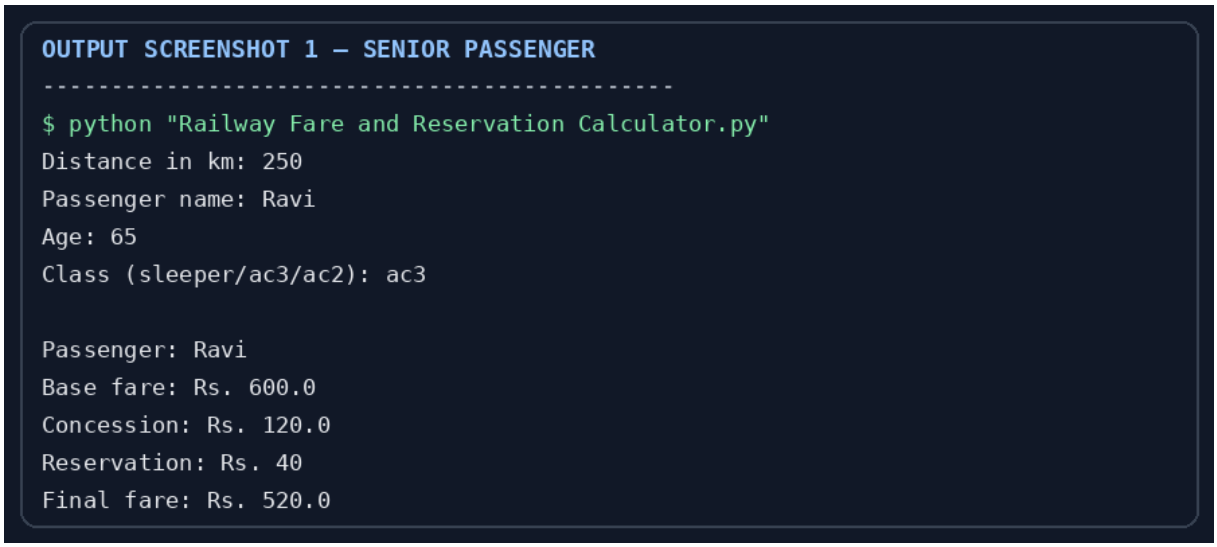
## 8. Test Cases

| Test | Input | Expected Output | Purpose |
|---|---|---|---|
| 1 | 250 km, Ravi, age 65, ac3 | Base Rs. 600; concession Rs. 120; reservation Rs. 40; final Rs. 520 | Senior passenger concession |
| 2 | 0 km, Ravi, age 30, sleeper | Invalid input. | Invalid distance |
| 3 | 100 km, Anu, age 25, sleeper | Base Rs. 120; concession Rs. 0; reservation Rs. 20; final Rs. 140 | Passenger below 60 |
| 4 | 50 km, Sam, age 60, | Base Rs. 180; | Age exactly 60 |

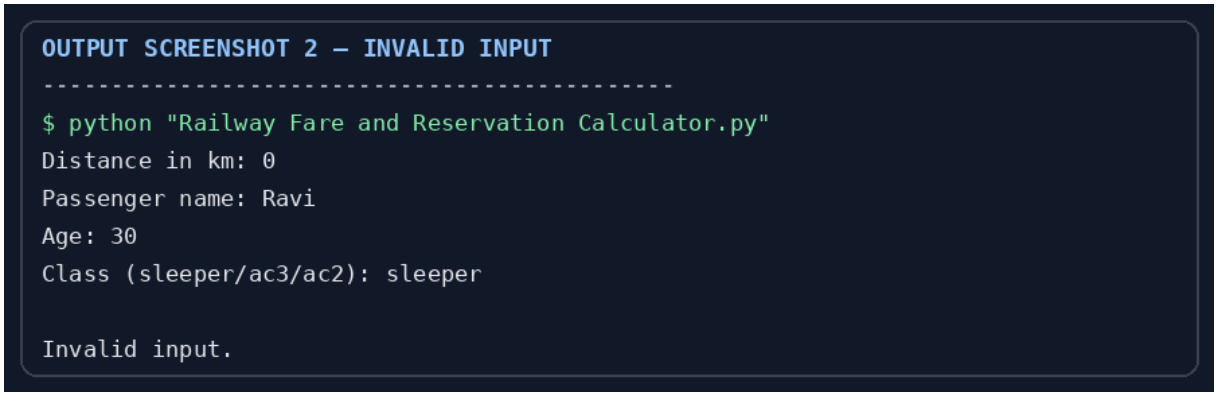
| | | | |
|---|---|---|---|
| | ac2 | concession Rs. 36; reservation Rs. 50; final Rs. 194 | |
| 5 | 100 km, Dev, age 30, ac1 | Invalid input. | Invalid travel class |

## 9. Output Screenshots

Screenshot 1: Senior passenger using AC 3-tier class.




Screenshot 2: Invalid distance.




Screenshot 3: Passenger below age 60.

```
OUTPUT SCREENSHOT 3 – NO CONCESSION
-------------------------------------------
$ python "Railway Fare and Reservation Calculator.py"
Distance in km: 100
Passenger name: Anu
Age: 25
Class (sleeper/ac3/ac2): sleeper

Passenger: Anu
Base fare: Rs. 120.0
Concession: Rs. 0
Reservation: Rs. 20
Final fare: Rs. 140.0
```

## 10. Conclusion

The Railway Fare and Reservation Calculator demonstrates input handling, dictionaries, conditional statements, arithmetic calculations, and rounding in Python. It calculates a class-based fare, applies the concession for passengers aged 60 or above, adds the reservation fee, and displays the final amount. Invalid distance, age, or class values result in an error message.

Limitation: The program does not validate non-numeric input for distance or age and does not perform an actual railway reservation.